a Declare roles once in the RAD v2 prelude

tag	type	role payload
b0	u32/u64	Barcode(level 0, Ls)
b1	u32/u64	Barcode(level 1, Lc)
u	u32/u64	Umi(Lu)

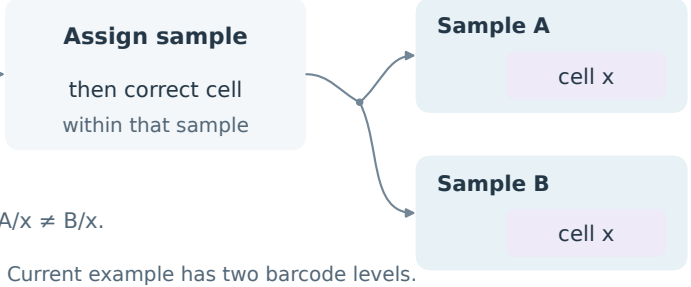
Level 0: sample

Level 1: cell

Role suffix: code:u8 | payload length:u8 | payload bytes. Ls, Lc, Lu: nucleotide lengths.

b Carry values; correct and group by sample + cell

n align	S	C	U	assignments
1	A	x	u1	P1
1	B	x	u1	P1



Schematic record excerpts; identities remain distinct: A/x ≠ B/x.

Two-bit-encoded values; schema is not repeated per read. Current example has two barcode levels.